

STEROID HORMONE RECEPTORS AND PHARMACOLOGY

J. P. RAYNAUD, M. M. BOUTON, M. MOGULEWSKY, T. OJASOO, D. PHILIBERT,
G. BECK*, F. LABRIET† and J. P. MORNON‡

Centre de Recherches Roussel-Uclaf, 93230 Romainville,

*Institut de Biologie Moléculaire et Cellulaire, 67000 Strasbourg, France

†Le Centre Hospitalier de l'Université Laval, Québec, G1V 4G2, Canada and

‡Laboratoire de Minéralogie et Cristallographie, Université de Paris VI, 75230 Paris, France

SUMMARY

The cumbersome nature of many pharmacological tests has seriously limited the full-scale study of the hormonal activity profile of a large number of molecules. To overcome this handicap, a screening system has been devised based on the fact that (a) interaction with the relevant hormone receptor is the key step in triggering a particular hormone response (e.g. estrogenic, progestomimetic, androgenic, mineralocorticoid or glucocorticoid) and (b) the kinetics of the interaction define the nature of the response, either agonistic or antagonistic. The data we have so far obtained with over 900 compounds have been classified according to binding specificity and kinetics and correlated with observations on the crystal-line or calculated conformations of chemically related compounds, in order to pave the way to new structure-activity relationships. However, although tailored compounds can thus be designed, their use as drugs depends, on the one hand, upon their pharmacokinetics and metabolism in man and, on the other, on receptor availability at the time of administration. At the present stage, highly specific ligands derived from this screening system are being used not only as drugs but as tools to study the physiological regulation of hormone receptors in both peripheral and central target tissues.

INTRODUCTION

The following retrospective of 5 years study on the relevance of steroid hormone receptors to pharmacology will not attempt to cover all the major advances in the field; this formidable task has been the aim of several excellent reviews [1-3]. Instead, we shall describe the line of thought we have pursued in our laboratory over these years and show how, by devising a routine screening system on the basis of available knowledge on steroid hormone receptor interactions, it has been possible to offer explanations for known pharmacological observations, to generalize observations derived from one class of hormones to others, and to select or design new compounds which can be efficiently used as tracers ("tags") for the further study of these interactions.

Full descriptions of the routine screening system devised over 5 years ago [4-6] are given in two recent papers [7, 8] to which the reader is referred for details. In brief, the competition of each newly synthesized compound, either routinely made available or tailor-made, for binding to the estrogen, progestin, androgen, mineralocorticoid and glucocorticoid receptors is measured in high-speed supernatants ("cytosols") prepared from receptor-rich target organs which are the end-points of classic biological assays (for subsequent structure-affinity-activity, correlations). The receptor is singled out with radiolabeled ligands, i.e. either with the natural hormone or, if necessary, with a potent synthetic agonist. Relative binding affinities (RBAs) are measured under two or

even three different sets of incubation conditions chosen in such a way that the variation in RBAs gives an indication of the kinetics of the interaction between the test-compound and the receptor [9-11].

In the present paper, three examples have been selected to illustrate the merits and repercussions of such a screening system: (a) the study of the specificity of binding of progestins currently used in the pill will show how binding to the androgen receptor can account for some of their known androgenic side-effects but also for part of their action at the central level [12, 13], (b) the study of the interaction kinetics of test-compounds for binding to the mineralocorticoid and glucocorticoid receptors will show how principles on anti-hormone action first deduced for estrogens [9, 14], then extended to progestins and androgens [11, 15, 16], have now been generalized to the corticoids, (c) the study of receptor regulation, e.g. of the variations in brain progestin receptor during facilitation and inhibition of lordosis [17-19], will illustrate new applications of known tags previously selected by this screening system [8, 20]. Furthermore, we will show how the disadvantages of some of these tags can now largely be overcome by using them in combination with ligands having complementary specificity profiles.

MATERIALS AND METHODS

Test-compounds

The test-compounds listed in Table 1 have been used.

Table 1. Chemical names of test-substances

Aldosterone	11 β ,21-Dihydroxy-3,20-dioxo-4-pregnen-18-al
Canrenone	17 β -Hydroxy-3-oxo-17 α -pregna-4,6-diene-21-carboxylic acid, γ -lactone
Chlormadinone acetate	17 α -Acetyloxy-6-chloro-pregna-4,6-diene-3,20-dione
Corticosterone	11 β ,21-Dihydroxy-4-pregnene-3,20-dione
Cortisol	11 β ,17 α ,21-Trihydroxy-4-pregnene-3,20-dione
Cortivazol (21-OH)	11 β ,17 α -Dihydroxy-6,16 α -dimethyl-2'-phenyl-2'H-pregna-2,4,6-trieno [3,2-c] pyrazol-20-one
Cyproterone acetate	17 α -Acetyloxy-6-chloro-1,2-dihydro-(1 β ,2 β)-3'H-cyclopropa-[1,2]pregna-1,4,6-triene-3,20-dione
Demegestone (RU 2453)	17 α -Methyl-19-nor-pregna-4,9-diene-3,20-dione
Desoxymetasone	9 α -Fluoro-11 β ,21-dihydroxy-16 α -methyl-pregna-1,4-diene-3,20-dione
Dexamethasone	9 α -Fluoro-11 β ,17 α ,21-trihydroxy-16 α -methyl-pregna-1,4-diene-3,20-dione
5 α -Dihydrotestosterone (5 α -DHT)	17 β -Hydroxy-5 α -androstan-3-one
Estradiol	1,3,5(10)-Estratriene-3,17 β -diol
Estriol	1,3,5(10)-Estratrien-3,16 α ,17 β -triol
Ethinodiol diacetate	19-Nor-4-pregnen-20-yne-3 β ,17 β -diol diacetate
Gestrinone (RU 2323)	13 β -Ethyl-17 β -hydroxy-18,19-dinor-pregna-4,9,11-trien-20-yn-3-one
Hydroxyflutamide	2-Hydroxy-2-methyl-N-[4-nitro-3-(trifluoromethyl)phenyl]-propanamide
Medroxyprogesterone acetate	17 α -Acetyloxy-6 α -methyl-4-pregnene-3,20-dione
Megestrol acetate	17 α -Acetyloxy-6-methyl-pregna-4,6-diene-3,20-dione
2-Methyl estradiol	2-Methyl-1,3,5(10)-estratriene-3,17 β -diol
Methyltrienolone (RU 1881)	17 β -Hydroxy-17 α -methyl-4,9,11-estratrien-3-one
Moxestrol (RU 2858)	11 β -Methoxy-19-nor-17 α -pregna-1,3,5(10)-trien-20-yne-3,17 β -diol
Norethindrone	17 β -Hydroxy-19-nor-17 α -pregn-4-en-20-yn-3-one
Norgestrel	13 β -Ethyl-17 β -hydroxy-18,19-dinor-17 α -pregn-4-en-20-yn-3-one
Norgestrienone (RU 2010)	17 β -Hydroxy-19-nor-17 α -pregna-4,9,11-trien-20-yn-3-one
Progesterone	4-Pregnene-3,20-dione
Promegestone (RU 5020)	17 α ,21-Dimethyl-19-nor-pregna-4,9-diene-3,20-dione
Prorenone	6,7-Dihydro-17 β -hydroxy-3-oxo-(6 α ,7 α) 3'H-cyclopropa [6,7] 17 α -pregna-4,6-diene-21-carboxylic acid, γ -lactone
RU 16117	11 α -Methoxy-19-nor-17 α -pregna-1,3,5(10)-trien-20-yne-3,17 β -diol
RU 23908	5,5-Dimethyl-3-[4-nitro-3-(trifluoromethyl)phenyl]-imidazolidine-2,4-dione
RU 24411	2,2-Dimethyl-21-hydroxy-19-norpregn-4-ene-3,20-dione
RU 26143	2-Ethynyl-5 α -A-norestrane-2,17 β -diol (isomer α)
RU 2956	17 β -Hydroxy-2,2,17 α -trimethyl-estra-4,9,11-trien-3-one
RU 2999	17 β -Hydroxy-17 α -methyl-2-oxa-estra-4,9,11-trien-3-one
Spirolactone	7 α -Acetylthio-17 β -hydroxy-3-oxo-4-pregnene-21-carboxylic acid, γ -lactone
Testosterone	17 β -Hydroxy-4-androsten-3-one
Triamcinolone acetate	9 α -Fluoro-11 β ,21-dihydroxy-16 α ,17 α -(1-methylethylidene) bis (oxy)]-pregna-1,4-diene-3,20-dione

Animals

The following were used: immature female mice (Swiss), adult male and female rats (Wistar and Sprague-Dawley) and immature female rabbits (Normandy and New Zealand).

Crystallographic studies

The crystalline and calculated conformations of progesterone and gestrinone were determined and/or analysed as previously described [21, 22].

Receptor binding studies

Relative binding affinity (RBA) determination. The relative binding affinities (RBAs) of the test-compounds for cytosolic steroid hormone receptors were determined by a dextran-coated charcoal (DCC) adsorption technique as previously described [8]. The species and organs for the preparation of homogenates and the incubation conditions are indicated in the table footnotes. The RBAs of the natural hormones, estradiol, progesterone, testosterone and aldosterone and of the synthetic hormone, dexamethasone, for the estrogen, progestin, androgen, mineralocorticoid and glucocorticoid receptors, respectively, were taken to be equal to 100.

Measurement of cytosolic progestin binding sites. Cytosolic progestin binding sites were measured in the uterus, pituitary and hypothalamus of ovariectomized adult female rats by a DCC adsorption technique as previously described [19]. In the sequential inhibition experiments, 24 h after priming with 2 µg estradiol benzoate (EB) or vehicle (sesame oil) only, the rats received either sesame oil, 5–50 µg promegestone or 0.5–5 mg progesterone. Progestin binding sites were measured 24 h later. In the concurrent inhibition experiments, the rats received either sesame oil, 2 µg EB alone, 2 µg EB plus 10–250 µg promegestone, or 2 µg EB plus 1–10 mg progesterone. Progestin binding sites were measured 42 h later. Results are expressed as the percentage inhibition of the EB-induced increase in progestin binding sites.

Cell cultures

Tyrosineaminotransferase (TAT) assay in hepatoma tissue culture (HTC) cells. The HTC cells were grown at 37°C as suspensions in modified SWIM'S S₇₇ medium containing 10% calf serum. The medium was buffered with 0.5 g NaHCO₃/l and 5 × 10⁻² M tricine adjusted to pH 7.4 at 37°C. In this medium, the cells have a generation time of 24–30 h and remain in the log phase of growth between a cell density of 2 to 8 × 10⁵ cells/ml. All incubations were carried out in spinner flasks at 37°C under magnetic stirring at a speed of 100 rev./min. Test-steroids were added in a 10 µl solution of ethanol per 10 ml of culture (density ~ 2 × 10⁵ cells/ml) and equivalent amounts of ethanol were always added to the control samples. Cell viability was tested by trypan blue exclusion [23]. To assay TAT, 2 ml samples of the cell culture were

removed and the cells were collected by centrifugation. The pellets were washed twice in isotonic saline and thereafter suspended and chilled in 1 ml of 5 × 10⁻² M potassium phosphate buffer (pH 7.6), 2 × 10⁻³ M 2-oxoglutarate 1 × 10⁻⁴ M pyridoxal phosphate and 1 × 10⁻³ M EDTA. The cells were disrupted with an ultra-sonicator (250 TS-20K, Annemasse, France) by 2 consecutive exposures to 70 volts for 20 s bursts. The enzyme was assayed at 37°C by the method of Diamondstone [24]. One unit of activity represents the formation of 1 µmole of *p*-OH-phenylpyruvate/min. Enzyme-specific activity is expressed as milliunits of TAT/mg of cell protein.

Biological assays

Progestational activity. (a) Clauberg test: Immature rabbits were primed with 5 µg/day of estradiol for 5 days before treatment either s.c. or orally with the test-compounds for a further 5-day period. The animals were killed 24 h after the last treatment, and the uteri were then excised and fixed in Bouin fluid. The endometrial proliferation was graded histologically in McPhail units. (b) McGinty test: After estrogen priming as above, the test-compounds were administered 3 days later locally (between 2 uterine ligatures) and the rabbits were killed 72 h after treatment. Endometrial proliferation was graded as above. (c) Deciduoma induction: Rats castrated on the day of estrus received the test-compounds by the s.c. route for 10 days. On the 5th day of treatment, a thread was passed through one uterine horn. The percentage of rats with deciduomata 24 h after the last treatment was recorded. (d) Pregnancy maintenance: Pregnant rats castrated on day 8 of pregnancy received the test-compounds by the s.c. route from day 3 to day 19 of pregnancy. The percentage of live foetuses was determined 24 h after the last treatment.

Androgenic (a) and anti-androgenic (b) activities (stimulation and inhibition of prostate weight) in castrated rats. (a) Castrated adult male rats received the test-compounds daily for 7 days by the s.c. route and were killed 24 h after the last injection. The increase in ventral prostate weight compared to untreated controls was determined. (b) Castrated adult male rats treated as above simultaneously received a total dose of 350 µg testosterone propionate and were killed 24 h after the last injection. The inhibition in the androgen-induced ventral prostate weight increase was recorded.

Inhibition of LH secretion in male (a) and female (b) rats. (a) Adult rats orchidectomized 2 weeks previously received s.c. injections of the test-compounds daily for 7 days before the measurement of plasma LH by double anti-body radioimmunoassay using rat hormones (LH-I-1, LH-RP-1) and rabbit antiserum (anti-LH-S-1) kindly supplied by the National Institute of Arthritis, Metabolism and Digestive Diseases. (b) Female 4-day cycling rats received s.c. injections of the test-compounds at 9.00 a.m. and 4.00 p.m. on the day of estrus. Plasma LH concentration was

Table 2. Relative binding affinities (RBAs) of a series of well-known progestins for cytosolic steroid hormone receptors

	PG 24 h 0°C	AND 2 h 0°C	MIN 30 min 25°C	GLU 4 h 0°C
Progesterone	100	5.5 ± 0.6	8.0 ± 1.0	0.2 ± 0.1
Chlormadinone acetate	321 ± 35	21 ± 4	<0.1	7.8 ± 0.9
Demegestone	420 ± 59	1.1 ± 0.4	1.4 ± 0.1	4.6 ± 0.2
Promegestone	533 ± 40	1.4 ± 0.4	0.3 ± 0.1	14 ± 2
Megestrol acetate	118 ± 10	19 ± 4	<0.1	3.9 ± 1.4
Medroxyprogesterone acetate	306 ± 25	52 ± 8	0.2 ± 0.07	0.14 ± 0.03
Norethindrone	263 ± 10	44 ± 3	0.2 ± 0.06	2.8 ± 0.1
Norethindrone acetate	64	1.6	0.1	1.3
Norgestrienone	47 ± 12	71 ± 13	1.8 ± 0.5	11 ± 3
Norgestrel	907 ± 184	87 ± 23	0.9 ± 0.3	14 ± 2
Gestrinone	48 ± 8	83 ± 3	4.2 ± 0.4	77 ± 16
Ethinodiol diacetate	2.2	0.2	<0.1	0.2

PG, AND, MIN and GLU = Progestin (rabbit uterus), androgen (rat prostate), mineralocorticoid (rat kidney) and glucocorticoid (rat liver) receptors, respectively. The RBAs of progesterone, testosterone, aldosterone and dexamethasone are taken to be equal to 100. The values are means ± SEM or the means of two determinations differing by less than 15%.

measured as above on samples obtained in the afternoon (4.00 p.m.) of expected proestrus.

Uterotrophic (a) and anti-estrogenic (b) activities. (a) Immature female mice received the test-compounds for 3 days by the s.c. route. They were killed 24 h after the last injection and their uterine weight was determined. (b) Immature female mice treated as above simultaneously received a total dose of 0.27 µg estradiol and were killed 24 h after the last injection. The inhibition in the estradiol-induced uterine weight increase was recorded.

RESULTS

1. Hormone-specificity of progestins

The binding of several progestins, both androstane and pregnane derivatives, currently used in the pill, to cytosolic progestin and androgen receptors is shown in Table 2. Binding to the progestin receptor was highest for the androstane derivative, norgestrel, and the pregnane derivatives, promegestone and demegestone, slightly less for the pregnane acetates, chlormadinone acetate and medroxyprogesterone acetate, and for norethindrone and considerably less for the remaining compounds. Binding to the androgen receptor was marked for the androstane derivatives such as norethindrone, norgestrel and the trienes, norgestrienone and gestrinone, and for the pregnane acetate, medroxyprogesterone acetate, considerably less for chlormadinone acetate and megestrol acetate and virtually nonexistent for the pregnane non-acetates (demegestone and promegestone). Trienes were furthermore characterized by their competition for binding to the corticoid receptors. It has been recently shown in a detailed study on the conformational mobility of a series of homologous steroid monoenes, dienes and trienes [22] that trienes are particularly flexible molecules. Conformational mobility has been correlated with receptor binding and the results suggest that high flexibility could account for the ability of a triene to bind to several receptors. Compared to progesterone which adopts three differ-

ent conformations in the crystal that are fairly easily superimposable (Fig. 1), the triene gestrinone adopts at least five conformations that cannot be superimposed; however, at least one of these conformations resembles that of progesterone, thus possibly explaining the ability of gestrinone to bind to the progestin receptor.

Binding of these progestins to the progestin and androgen receptors accounts for their progestomimetic and androgenic activities, respectively. All the compounds with higher RBAs than progesterone were more active than progesterone as illustrated by the results of the Clauberg test by the s.c. route (Fig. 2a). The plateau-shaped curve recorded in the case of gestrinone was explained by the fact that this compound is an impeded progestin, known to have some anti-progestational activity [25]. The compounds were classified slightly differently after administration by the oral route on account of differences in absorption and metabolism; in particular, progesterone was inactive by the oral route (Fig. 2b). Furthermore, whereas all the pregnane derivatives were also highly active in other progestational activity tests such as endometrial proliferation on local administration, pregnancy maintenance, and deciduoma induction (Table 3), this was not the case for androstane derivatives which, except for norgestrel, were inactive.

Figures 3 and 4 confirm that the androgenic activity of androstane progestins is in agreement with their binding to the androgen receptor. Androstane, but not pregnane, progestins increased ventral prostate weight in orchidectomized rats (Fig. 3). Androgens are known to inhibit LHRH-induced LH release in anterior pituitary cells in culture [26] and progestins such as medroxyprogesterone acetate and norgestrel, which have androgenic activity, exert a similar effect [12]. Norgestrel and other androgenic progestins also inhibited plasma LH levels *in vivo* in orchidectomized rats (Fig. 4).

The androgen binding of the androstane derivatives can also be held responsible for their slight uterotro-

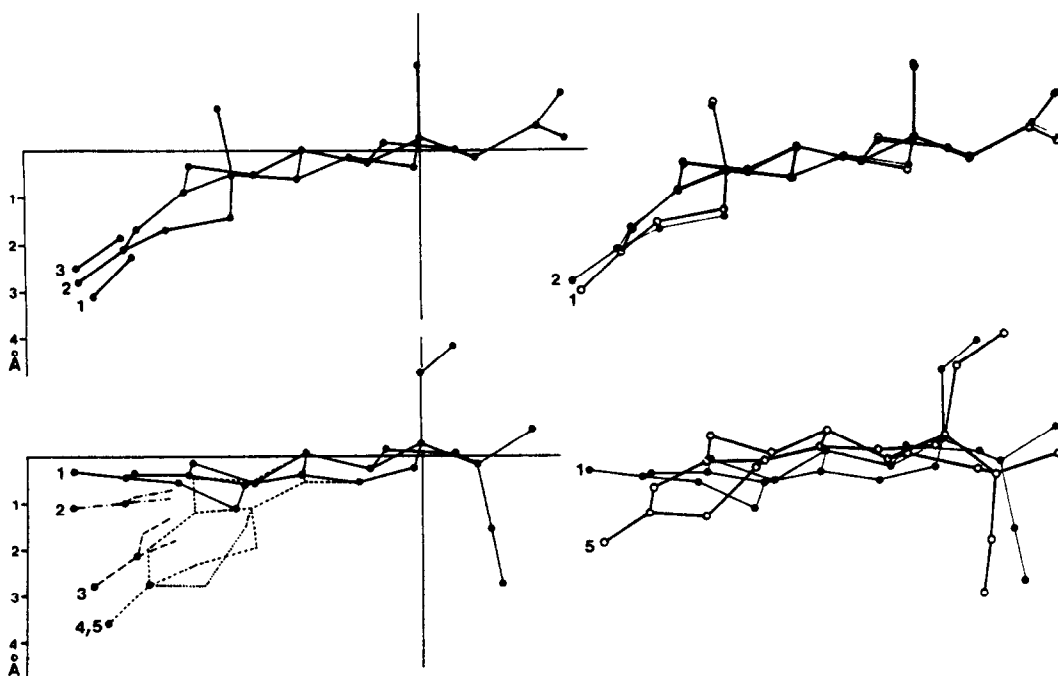


Fig. 1. Crystalline conformations of progesterone (top) and of gestrinone (bottom). The left-hand diagrams illustrate the superimposition of conformations encountered in the crystal using the D-ring reference system which reveals differences in molecular curvature; the right-hand diagrams illustrate the superimposition of conformations 1 and 2 of progesterone and 1 and 5 of gestrinone by a best-fit programme which reveals differences in overall conformation [22].

phic activity in the mouse (Fig. 5a). Pregnane derivatives, on the other hand, had virtually no effect on uterine weight. However, although under our routine screening conditions (incubation for 2 h at 0°C), none of the progestins, whether androstane or pregnane derivatives, competed for estrogen binding [7, 8], on changing these conditions (very short incubation times at 0°C), some of the progestins did actually compete [27]. The slight uterotrophic effect might also be related to this competition for estrogen receptor binding.

Androstane and pregnane progestins, however, had two fundamental points in common. Both had anti-estrogenic activity as measured by the inhibition of the estradiol-induced increase in uterine weight (Fig. 5b). This activity was most marked for the androstane derivative, norgestrel, and the pregnane derivative, promegestone. Both were highly potent inhibitors of the spontaneous LH surge in the female cycling rat (Fig. 6). Promegestone was as active as norgestrel but slightly less active than gestrinone.

II. Interaction kinetics and anti-hormonal activity

According to previous studies [9–11], a certain class of anti-estrogens would appear to exert their antagonistic activity by virtue of the fast dissociation of the complex they form with the estrogen receptor compared to that of the natural hormone. The dissociation of this complex can be measured either directly with radiolabeled hormone or indirectly by

comparing RBAs recorded under different incubation conditions [10, 11]. For instance, Table 4 shows that whereas the potent estrogen moxestrol forms a slowly-dissociating receptor complex (the RBA increases from 12 to 122 on increasing incubation time), the following compounds: 2-methylestradiol, RU 16117, estriol and RU 26143 form fast-dissociating complexes (the RBAs decrease with incubation time). As indicated in Fig. 7, all these compounds which form fast-dissociating complexes have anti-estrogenic activity. Similarly, steroids such as RU 2956 and cyproterone acetate and non-steroids such as hydroxyflutamide and RU 23908, which form fast-dissociating complexes with the androgen receptor (Table 5), are potent anti-androgens (Fig. 8).

Measurements of binding to the mineralocorticoid and glucocorticoid receptors have so far been severely handicapped by the lack of specific ligands for these receptors, in particular for the mineralocorticoid receptor. All available radioligands (e.g. labeled aldosterone, dexamethasone, 9 α -fluorocortisol etc.) bind to a greater or lesser extent to both these receptors and, in most target tissues, both receptors are present concurrently making the distinction even more difficult. Exceptions are, for instance, the cytosolic mineralocorticoid receptor in frog skin [28]; however, the specificity of this receptor would appear to be somewhat different from that of the mineralocorticoid receptor in rat tissues insofar as 9 α -fluorocortisol does not bind to the frog skin receptor but binds to the rat receptor [7]. For this reason, in the study of

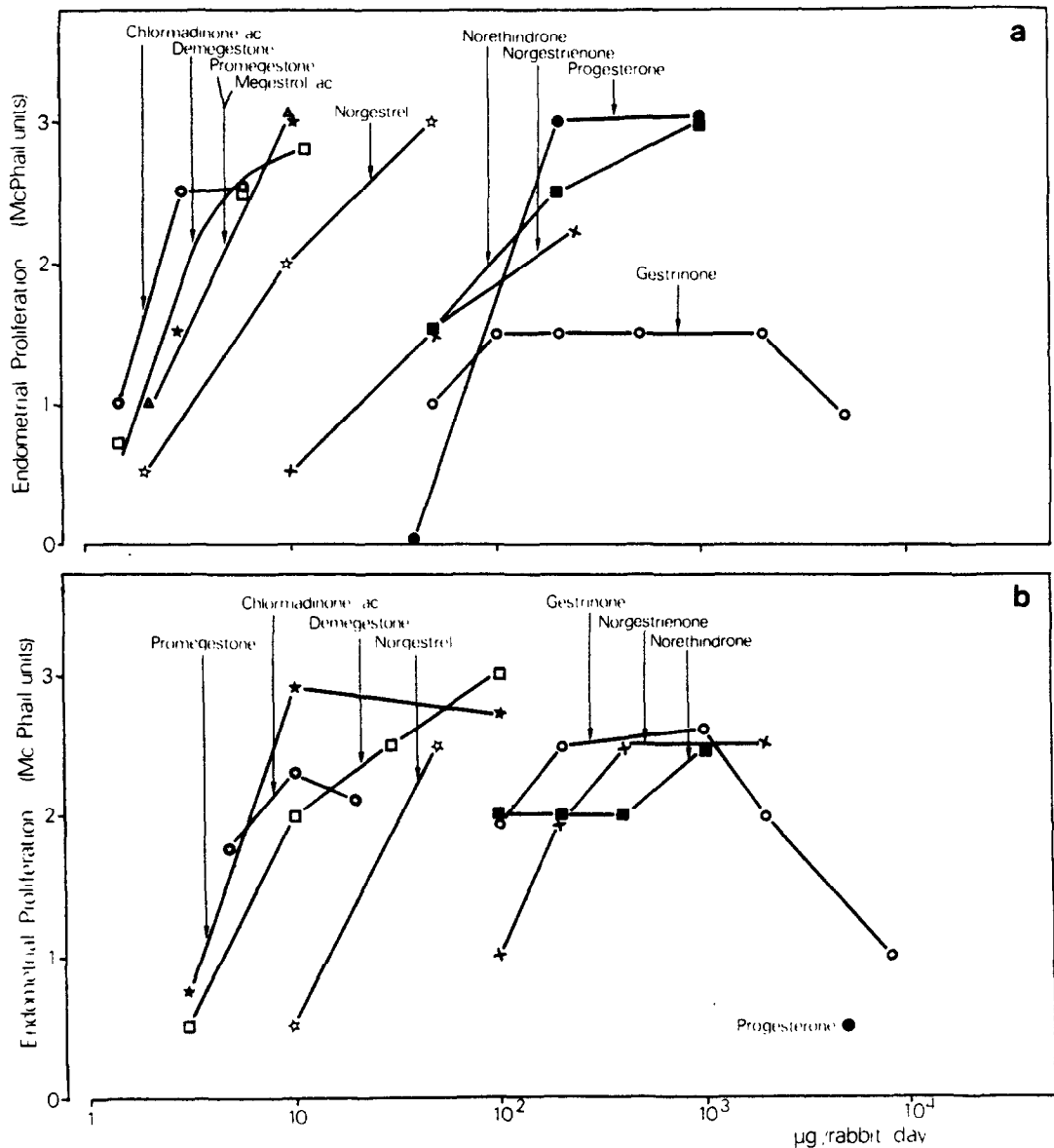


Fig. 2. Progesterational activity of progestins in the Clauberg test. (a) S.c. route. (b) Oral route.

Table 3. Progesterational activity

	Endometrial proliferation*	Pregnancy maintenance	Deciduoma induction
Progesterone	++	++	++
Chlormadinone acetate	+++	+++	+++
Demegestone	+++	+++	+++
Promegestone	+++	+++	+++
Megestrol acetate	+++	+++	+++
Medroxyprogesterone acetate	+++	++	++
Norethindrone	—	—	—
Norethindrone acetate	—	—	—
Norgestrienone	—	—	—
Norgestrel	(+)	+	(+)
Gestrinone	—	—	—
Ethinodiol diacetate	—	—	(+)

Potency has been graded on a scale from (+) to + + +. Inactive = —.
* Local administration.

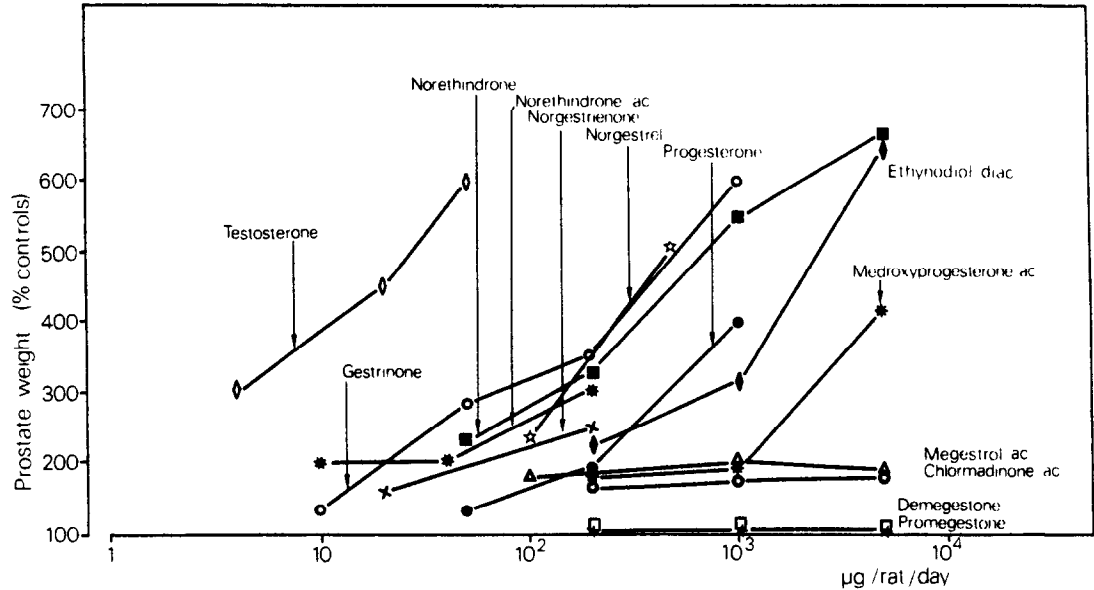


Fig. 3. Androgenic activity (s.c.) of progestins in the rat.

the interaction kinetics of corticoids, we have measured glucocorticoid binding with labeled dexamethasone (in spite of its slight mineralocorticoid binding) and mineralocorticoid binding with labeled aldosterone in the presence of an excess of an unlabeled highly specific glucocorticoid, RU 26988, to

mask binding to the glucocorticoid receptor in rat kidney (for further details on this compound and the technique used, see below and p. 309 of this volume). The results are given in Tables 6 and 7 which confirm the well-known anti-mineralocorticoid properties of prorenone, spironolactone and its active

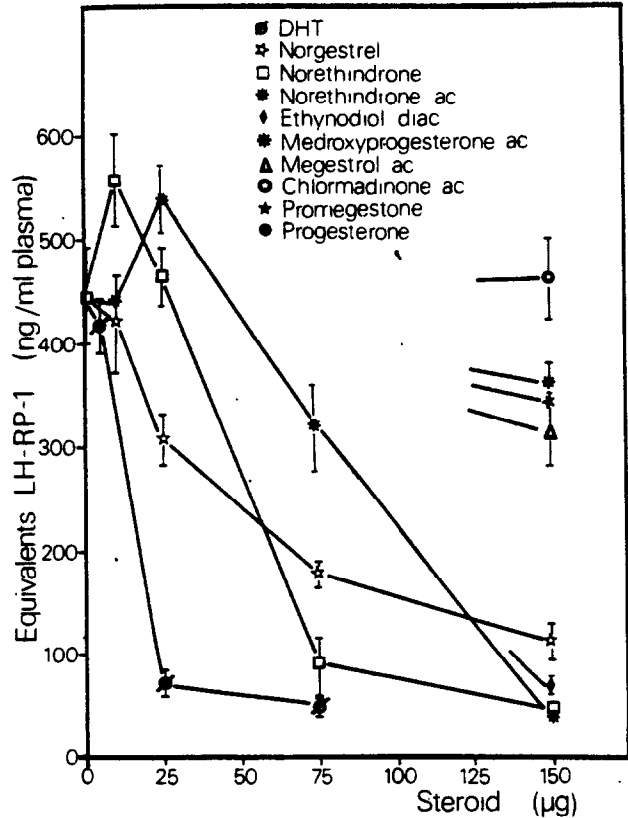


Fig. 4. Effect of injected progestins on plasma LH levels in orchidectomized rats.

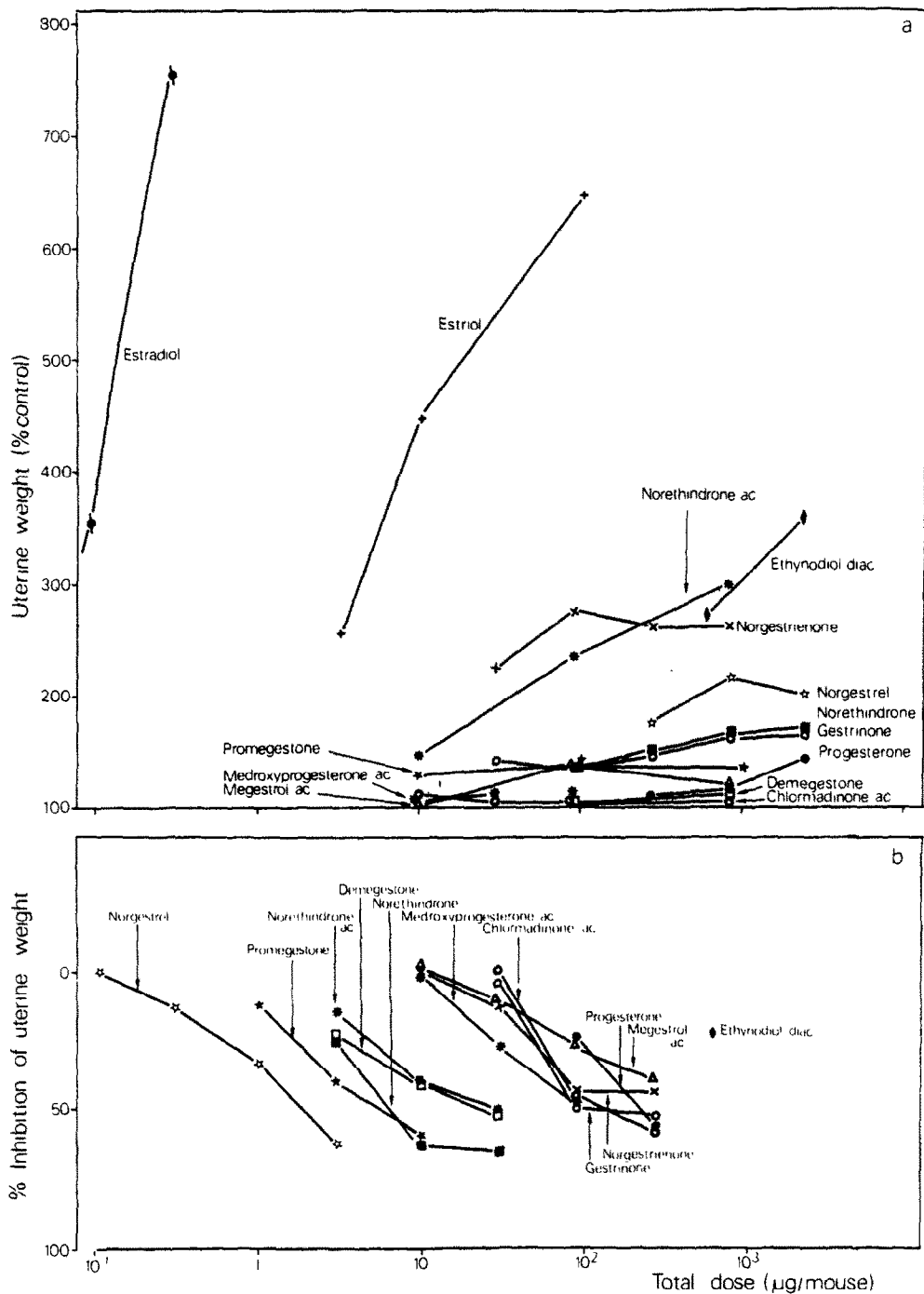


Fig. 5. (a) Uterotrophic activity (s.c.) of progestins in the mouse. (b) Anti-estrogenic activity (s.c.) of progestins in the mouse.

metabolite, canrenone, and of the new derivative RU 24411. All these compounds have decreasing RBAs on increasing incubation time. On the basis of these binding studies, the natural hormone progesterone would seem to have both anti-mineralocorticoid (Table 6) and anti-glucocorticoid (Table 7) properties; these have indeed been reported in the literature [29–31]. Two compounds with increasing RBAs have been included in Table 7; these are the potent gluco-

corticoids, 21-OH-cortivazol and desoximetasone. Their ability to induce tyrosineaminotransferase activity in hepatoma tissue culture cells, compared to that of dexamethasone, is illustrated in Fig. 9. Compounds with decreasing RBAs (i.e. RU 2999, progesterone and promegestone) were inactive in this test, but as shown in Fig. 10 were able to counteract the enzyme activity induced by the potent glucocorticoids. In conclusion, in the absence of any metabolizing

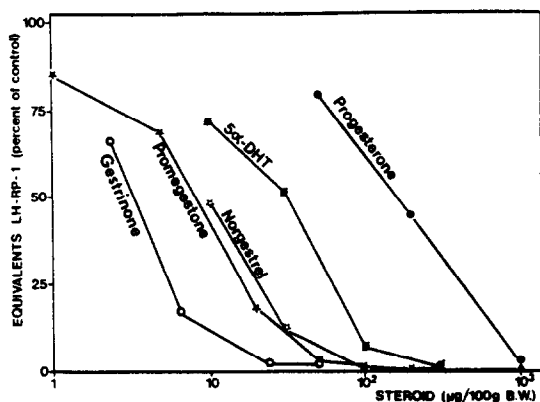


Fig. 6. Effect of progestins injected at 9 a.m. and 4 p.m. on estrus on the plasma LH concentration measured in the afternoon (4 p.m.) of expected proestrus in 4-day cycling rats.

activity in the cytosol and of a high level of plasma contamination introducing specific plasma proteins to which the various ligands may bind, the decrease in relative binding affinities can select anti-hormones whatever the hormonal class under study.

III. The application of steroid receptor "tags"

The detection and assay of steroid hormone receptors necessitates the use of radioligands which fulfil the following criteria: (a) minimal binding to steroid-specific plasma proteins, (b) no degradation on *in vitro* incubation with cytosol, (c) high radioactive labeling, yet low non-specific binding, (d) slower dissociation rate from the receptor than that of the endogenous hormone, (e) exclusive binding to the receptor of a single hormone class. Since the natural hormones themselves do not always fulfil all these criteria (e.g. estradiol and testosterone bind to contaminating sex-steroid binding protein in cytosol, progesterone binds to corticosteroid binding globulin, dihydrotestosterone is degraded on *in vitro* incubation even at 0°C, and progesterone dissociates extremely fast from its receptor) synthetic radioligands without these disadvantages are often used. These can be chosen on the basis of data from the routine screening system which indicate whether they meet the requirements for high specificity and suitable interaction kinetics. For instance, as indicated in Table 2, promegestone (RU 5020) binds with high affinity virtually exclusively to the progestin receptor and forms a slowly-

Table 5. Relative binding affinities for the cytosolic androgen receptor in rat prostate

	30 min 0°C	2 h 0°C	24 h 0°C
RU 1881	158 ± 26	203 ± 5	294 ± 26
5α-DHT	94 ± 10	120 ± 3	180 ± 39
Testosterone	100	100	100
RU 2956	55 ± 13	14 ± 2	3.6 ± 0.5
Cyproterone acetate	51	16 ± 1	5.4
Hydroxyflutamide	8.2	1.1	0.2
RU 23908	9.1	1.1	0.4

Mean ± SEM or mean of two determinations differing by less than 15%.

dissociating complex with this receptor; its RBA after 24 h incubation is 533 compared to 222 after only 2 h. Similarly, moxestrol (RU 2858) is specific to the estrogen receptor [7, 8] and forms a slowly-dissociating complex (see Table 4). As previously established [20, 32, 33], these compounds fulfil most of the above criteria (neither RU 5020 nor RU 2858 bind to specific plasma proteins and both are metabolically stable) and are now in routine use to detect and assay progestin and estrogen receptors by exchange in animal and human tissue specimens under different physiological and pathological conditions (e.g. during the cycle and during gestation [32], in breast cancer, endometrial cancer, prostate cancer [32, 34]).

These radioligands have also been efficiently used to detect brain steroid receptors and to draw correlations between the presence of these receptors and sexual behaviour. In particular, with the help of moxestrol, which does not bind to estradiol binding protein present in high concentrations in the plasma

Table 6. Relative binding affinities for the cytosolic mineralocorticoid receptor in rat kidney

	1 h 0°C	4 h 0°C	20 h 0°C
Aldosterone	100	100	100
Progesterone	183 ± 46	83 ± 11	24 ± 5
Spironolactone	182 ± 20	95 ± 10	7 ± 3
Canrenone	145	84	29
RU 24411	64 ± 4	10 ± 2	2.0 ± 0.5
	15 ± 2	6 ± 1	1.6 ± 0.7

Mean ± SEM.

Mineralocorticoid binding was measured in the presence of a specific glucocorticoid, RU 26988, in order to eliminate interference of binding to the glucocorticoid receptor in rat kidney.

Table 4. Relative binding affinities for the cytosolic estrogen receptor in immature mouse uterus

	2 h 0°C	5 h 25°C
Estradiol	100	100
Moxestrol	12 ± 1	122 ± 11
2-Methyl estradiol	41 ± 6	7.1 ± 1.3
Estriol	15 ± 2	4.3 ± 0.5
RU 16117	13 ± 1	4.0 ± 0.5
RU 26143	12 ± 1	2.1 ± 0.5

Mean ± SEM.

Table 7. Relative binding affinities for the cytosolic glucocorticoid receptor in hepatoma tissue culture cells

	4 h 0°C	20 h 0°C
Cortivazol (21-OH)	60 ± 10	90 ± 18
Desoximetasone	283 ± 22	310 ± 50
Dexamethasone	100	100
RU 2999	42 ± 9	19 ± 4
Promegestone	24 ± 6	9 ± 3
Progesterone	12 ± 2	2.0 ± 0.6

Mean ± SEM.

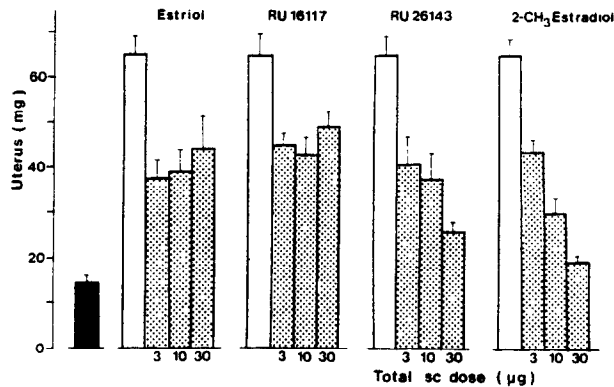


Fig. 7. Anti-estrogenic activity (s.c.) in the mouse (■ controls, □ 0.3 µg estradiol, ▨ 0.3 µg estradiol + anti-estrogen).

of the fetus and young rat, it has been possible to demonstrate specific radioactivity uptake in brain nuclei [35, 36]. The estrogen receptor is present in several target tissues (uterus, cortex, pituitary and hypothalamus) as early as the day of birth and is already functional (i.e. can induce progesterin binding sites) 4 days after birth [37, 38]. The presence of this functional estrogen receptor early in development could be linked with numerous studies on the effects

of estrogens on brain sexual differentiation. With the help of promegestone, which stabilises the progesterin receptor, it has been possible to measure the variations in the levels of progesterin receptor in the hypothalamus and to correlate them with observations on sexual behaviour. Progesterin receptor concentration is maximum between 40 and 48 h after estrogen priming at a time when the administration of progesterone greatly facilitates lordosis [19]. The administration of

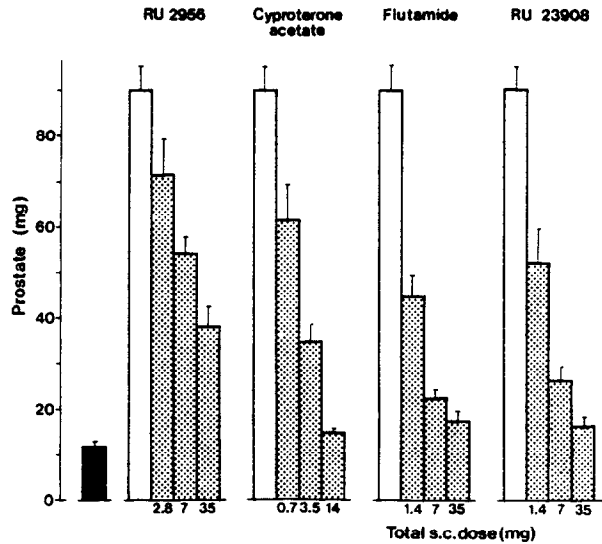


Fig. 8. Anti-androgenic activity (s.c.) in the rat (■ controls, □ 350 µg testosterone propionate, ▨ testosterone propionate + anti-androgen).

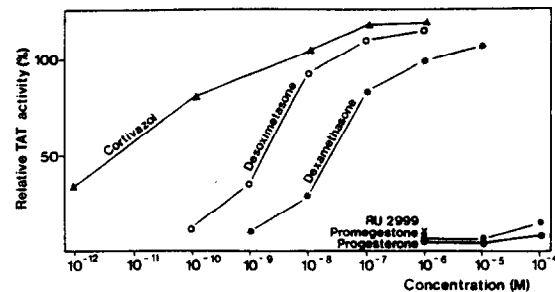


Fig. 9. Induction of tyrosineaminotransferase activity in hepatoma tissue culture cells.

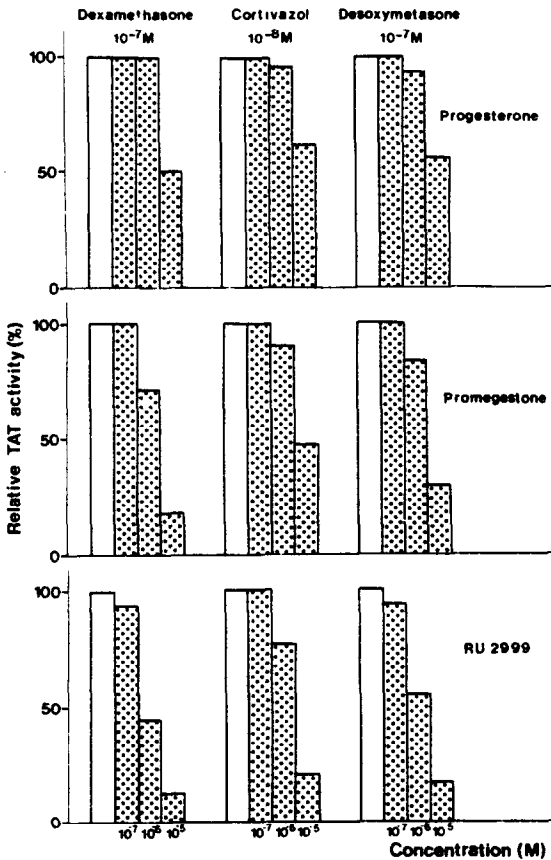


Fig. 10. Inhibition by three anti-glucocorticoids (progesterone, promegestone, RU 2999) of the tyrosineaminotransferase activity induced by three potent glucocorticoids (dexamethasone, cortivazol (acetate), desoxymetasone).

progesterone simultaneously with the estrogen (concurrent administration) or 24 h after estrogen administration (sequential administration) leads to a decreased progesterin receptor level in the hypothalamus 40 h after priming and the facilitative dose of pro-

gesterone can no longer induce lordosis [39]. As shown in Fig. 11, this inhibition of estrogen-induced progesterin binding sites by progesterone or promegestone parallels their inhibition of lordosis. Whether in sequential or concurrent inhibition experiments, the activity ratio between promegestone and progesterone was about 50 to 100.

These assays have been made possible by the fact that both moxestrol and promegestone fulfil well most of the criteria for tags including the specificity criterion. On the other hand, the tag, methyltrienolone (RU 1881), a potent androgen, which possesses the great advantage over dihydrotestosterone of binding minimally to sex-steroid binding protein under our exchange assay conditions and of being stable in the incubation medium, does not have complete steroid specificity. It binds equally to the androgen and progesterin receptors [7, 8, 34], a property which has incidentally been at the root of the discovery of a progesterin-like receptor in samples of human hyperplastic prostate and prostate adenocarcinoma [40]. RU 1881 can therefore only be used to assay androgen receptors in the presence of an excess of a ligand which binds to the progesterin receptor without competing for the androgen receptor. According to our screening data shown in Table 8, triamcinolone acetonide would seem to be a suitable candidate for this purpose, in particular since, as a result of its mineralocorticoid and glucocorticoid binding, it would also eliminate any interference, however minimal, of RU 1881 binding to these receptors. The combined use of labeled RU 1881 and radioinert triamcinolone acetonide for receptor assay has been described [34, 41, 42].

As mentioned above, assays of the mineralocorticoid and glucocorticoid receptors have been difficult for similar reasons; available radioligands such as labeled aldosterone and dexamethasone bind to both receptors (Table 8). To solve this problem, a strategy similar to the one used for the assay of androgen

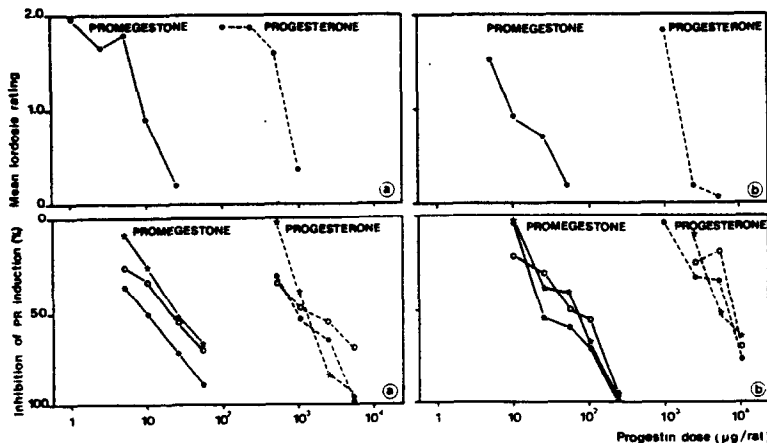


Fig. 11. Inhibition by promegestone and progesterone of lordosis and of the estrogen-induced increase in progesterin receptor in the hypothalamus (○), pituitary (●) and uterus (●). (a) Sequential inhibition (EB + sesame oil group = 0% inhibition, sesame oil group = 100% inhibition); (b) Concurrent inhibition (EB group = 0% inhibition, sesame oil group = 100% inhibition). The data on lordosis inhibition [39] are reproduced with the kind permission of Dr. J. D. Blaustein.

Table 8. Relative binding affinities in a routine screening system

	ES 2 h 0°C	PG 2 h 0°C	AND 2 h 0°C	MIN 30 min 25°C	GLU 4 h 0°C
Progesterone	<0.1	100	5.5 ± 0.6	8.0 ± 1.0	42 ± 3
5 α -DHT	<0.1	1.4 ± 0.7	120 ± 3	0.15	0.3
RU 1881	<0.1	208 ± 18	203 ± 5	18 ± 2	10
Triamcinolone acetoneide	<0.1	15 ± 4	<0.1	9 ± 2	141 ± 20
Cortisol	<0.1	<0.1	<0.1	14 ± 1	31 ± 4
Corticosterone	<0.1	5.1 ± 1.0	0.5 ± 0.2	33 ± 5	67 ± 2
Dexamethasone	<0.1	0.4 ± 0.1	<0.1	17 ± 2	100
Aldosterone	<0.1	1.1 ± 0.2	<0.1	100	14 ± 1
RU 26988	<0.1	0.3	<0.1	0.5 ± 0.1	127 ± 11

ES, PG, AND, MIN and GLU = Estrogen (mouse uterus), progestin (rabbit uterus), androgen (rat prostate), mineralocorticoid (rat kidney) and glucocorticoid (rat thymus) receptors respectively.

Mean ± SEM or mean of two determinations differing by less than 15%.

receptor can be used. In order to eliminate aldosterone binding to the glucocorticoid receptor, it is possible to add a highly specific glucocorticoid such as RU 26988 which binds to the glucocorticoid receptor only (Table 8). The physicochemical characteristics of RU 26988 binding and examples of its use are given elsewhere (this volume p. 309).

DISCUSSION

The implications of the many topics raised in this paper are too numerous to be effectively covered in a simple discussion. At a first glance, there seems to be a wide step from the side-effects of the progestins currently used in the pill to the selection of anti-hormones on the basis of interaction kinetics, from the detection of estrogen and androgen receptors in prostate adenocarcinoma to the correlation of brain estrogen and progestin receptor concentrations with sexual differentiation and behavior (for detailed discussions, see [7, 11, 19, 34]). And yet, all these studies have been greatly facilitated or even made possible by the routine screening system which has been set up to draw rapid and, in spite of the shortcomings of the system, relatively accurate profiles of over 900 molecules. The interpretation of the screening results and the problems which have cropped up en route have in fact frequently led to new openings in many fields. The lack of specificity of RU 1881 has led to the detection of the existence of a progestin-like receptor in benign hyperplastic prostates and prostate adenocarcinomas [40]; the lack of specificity of trienes compared to the corresponding dienes and monoenes [22] has stressed the importance of molecular flexibility besides receptor flexibility in binding to the receptors of several hormone classes. It can be hypothesized, for instance, that the conformation of the triene RU 1881 that stabilizes the receptor most efficiently is the one which approximates the conformation of a natural androgen rather than progestin, thus explaining that, in spite of its high affinity for both the androgen and progestin receptors in cytosol, *in vivo* the compound is a more potent androgen than progestin. The problem of defining appropriate incubation conditions for

such a routine screening system has revealed the necessity of choosing two different sets of incubation conditions to determine the relative life-span of a complex compared to that of the endogenous hormone complex.

Admittedly, a routine screening system lacks, by definition, the finesse of the experiment designed to uncover the more subtle aspects of the action of a particular molecule but, if it is accepted that the receptor interaction is the key-step in hormone action and that unifying principles underlie the action of all classes of steroid hormones, such a system can, if well-designed, efficiently separate the wheat from the chaff. We have already shown that differences in administration schedules and species can affect activity [11, 43] largely by modifying pharmacokinetics. That is why we have chosen for our screening system the receptors present in the target tissues used as endpoints in classical biological activity tests; our biochemical results can thus be extrapolated to the corresponding pharmacological tests and constitute an efficient, rapid and rational means of selection of compounds for further study. Moreover, it is hoped that attempts at correlating available data on the structure-affinity-activity of these 900 molecules will give rise to new leads in the mapping of the receptors corresponding to each hormone class in a biochemical environment (105,000 *g* supernatant) and to the design of new tailor-made drugs and tags.

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REFERENCES

1. Chan L. and O'Malley B. W.: Mechanism of action of the sex steroid hormones. *New Engl. J. Med.* **294** (1976) 1322-1328 & 1372-1381 & 1430-1437.
2. Gorski J. and Gannon F.: Current models of steroid hormone action: a critique. *Ann. Rev. Physiol.* **38** (1976) 425-450.
3. Higgins S. J. and Gehring U.: Molecular mechanisms of steroid hormone action. *Adv. Cancer Res.* **28** (1978) 313-397.

4. Raynaud J. P., Bonne C., Bouton M. M., Moguilewsky M., Philibert D. and Azadian-Boulanger G.: Screening for anti-hormones by receptor studies. *J. steroid Biochem.* **6** (1975) 615-622.
5. Raynaud J. P.: Une stratégie de recherche pour les hormones de synthèse: leurs interactions avec les récepteurs hormonaux. In: *Actualités Pharmacologiques* (Edited by Cheymol J., Boissier J. R. and Lechat P.), Masson & Cie, Paris, Vol. 29 (1977), pp. 49-64.
6. Raynaud J. P.: A strategy for the design of potent hormones. In *Medicinal Chemistry*, Vol. V (Edited by Mathieu J.), Elsevier, Amsterdam (1977) pp. 451-456.
7. Raynaud J. P., Ojasoo T., Bouton M. M. and Philibert D.: Receptor binding as a tool in the development of new bioactive steroids. In *Drug Design* (Edited by Ariens E. J.), Academic Press, New York, Vol. VIII (1979) pp. 169-214.
8. Ojasoo T. and Raynaud J. P.: Unique steroid congeners for receptor studies. *Cancer Res.* **38** (1978) 4186-4198.
9. Bouton M. M. and Raynaud J. P.: The relevance of kinetic parameters in the determination of specific binding to the estrogen receptor. *J. steroid Biochem.* **9** (1978) 9-15.
10. Bouton M. M., Bonne C. and Raynaud J. P.: "In vitro" screening for anti-hormones. *J. steroid Biochem.* **9** (1978) 836.
11. Raynaud J. P.: The mechanism of action of anti-hormones. In *Advances in Pharmacology and Therapeutics Vol. 1, Receptors* (Edited by Jacob J.), Pergamon Press, Oxford (1978) pp. 259-278.
12. Labrie F., Ferland L., Lagacé L., Drouin J., Asselin J., Azadian-Boulanger G. and Raynaud J. P.: High inhibitory activity of R 5020, a pure progestin, at the hypothalamo-adenohypophyseal level on gonadotropin secretion. *Fert. Steril.* **28** (1977) 1104-1112.
13. Raynaud J. P. and Bercovici J. P.: Introduction à la pharmacologie des progestatifs de synthèse. *Ann. Endocr. (Paris)* **40** (1979) 300-320.
14. Bouton M. M. and Raynaud J. P.: Impaired nuclear translocation and regulation: a possible explanation of anti-estrogenic activity. In *Research on Steroids* (Edited by Vermeulen A., Kloppe F., Sciarra F., Jungblut P. and Lerner L.), Elsevier, North-Holland Biomedical Press, Vol. 7 (1977) pp. 127-136.
15. Raynaud J. P., Bonne C. and Lagacé L.: On the mechanism of action of non-steroidal anti-androgens. *Endocrinology* **102** (suppl.) (1978) 304.
16. Raynaud J. P., Bonne C., Bouton M. M., Kelly P. and Lagacé L.: Action of a non-steroid anti-androgen, RU 23908, in peripheral and central tissues. *J. steroid Biochem.* **11** (1979) 93-99.
17. Moguilewsky M. and Raynaud J. P.: Progestin binding sites in the rat hypothalamus, pituitary and uterus. *Steroids* **30** (1977) 99-109.
18. Moguilewsky M. and Raynaud J. P.: Estrogen-sensitive progestin binding sites in the female rat brain and pituitary. *Brain Res.* (1979) **164** 165-175.
19. Moguilewsky M. and Raynaud J. P.: The relevance of hypothalamic and hypophyseal progestin receptor regulation in the induction and inhibition of sexual behaviour in the female rat. *Endocrinology* **105** (1979) 516-522.
20. Raynaud J. P., Ojasoo T. and Vaché, V.: Unusual steroids in measuring steroid receptors. In *Steroid Receptors and the Management of Cancer* (Edited by Thompson E. B. and Lippman M. E.), CRC Press, Cleveland (1979) in press.
21. Mornon J. P., Delettre J., Lepicard G., Bally R., Surcouf E. and Bondot P.: Interactions of hormonal steroids: progestogens. *J. steroid Biochem.* **8** (1977) 51-62.
22. Delettre J., Mornon J. P., Lepicard G., Ojasoo T. and Raynaud J. P.: Steroid flexibility and receptor specificity. *J. steroid Biochem.* (1979) in press.
23. Phillips H. J.: The exclusion tests for cell variability. In *Tissue Culture Methods and Applications* (Edited by Kruse P. S. and Patterson N. K.), Academic Press, New York (1973) pp. 406-408.
24. Diamondstone T. I.: Assay of tyrosine transaminase activity by conversion of p-hydroxyphenylpyruvate to p-hydroxybenzaldehyde. *Analyt. Biochem.* **16** (1966) 395-401.
25. Sakiz E., Azadian-Boulanger G. and Raynaud J. P.: Antiestrogens, antiprogesterones. In *Proceedings, VI International Congress of Endocrinology*, Washington, 1972, Excerpta Medica, Amsterdam. ICS 273 (1974) pp. 988-994.
26. Drouin J. and Labrie F.: Selective effect of androgens on LH and FSH release in anterior pituitary cells in culture. *Endocrinology* **98** (1976) 1528-1534.
27. Bouton M. M., Martin P. M. and Raynaud J. P.: The anti-estrogenic activity of progestins. *Cancer Treatment Reports* **63** (1979) 1151, no. 32.
28. Moguilewsky M. and Raynaud J. P.: Un modèle d'étude des antiminérocorticoïdes: la peau de grenouille. *J. Pharmacol. (Paris)* **7** (1976) 211-222.
29. Lan N., Matulich D. T., Graham B. and Baxter J. D.: Relative activities of steroids for binding to glucocorticoid and mineralocorticoid receptors: possible implications for anti-mineralocorticoid actions of the glucocorticoids. In *Endocrine Soc. Abstracts*, no. 284 (1978) 216.
30. Wambach G. and Higgins J. R.: Anti-mineralocorticoid action of progesterone in the rat: correlation of the effect on electrolyte excretion and interaction with renal mineralocorticoid receptors. *Endocrinology* **102** (1978) 1686-1693.
31. Rousseau G. G., Baxter J. D., Higgins S. J. and Tomkins G. M.: Steroid-induced nuclear binding of glucocorticoid receptors in intact hepatoma cells. *J. molec. Biol.* **79** (1973) 539-554.
32. McGuire W. L., Raynaud J. P. and Baulieu E. E.: (Eds.) *Progesterone Receptors in Normal and Neoplastic Tissues. Prog. Cancer Res. Ther.*, Vol. 4. Raven Press, New York (1977).
33. Raynaud J. P., Martin P., Bouton M. M. and Ojasoo T.: 11 β -Methoxy-17-ethynyl-1,3,5(10)-estratriene-3,17 β -diol (Moxestrol), a tag for estrogen receptor binding sites in human tissues. *Cancer Res.* **38** (1978) 3044-3050.
34. Raynaud J. P., Bouton M. M. and Martin P. M.: Steroid prostate adenocarcinoma: steroid hormone receptor assays and therapy. In *Steroid Receptors, Metabolism and Prostatic Cancer*, Excerpta Medica, Amsterdam (1980) in press.
35. Raynaud J. P.: Estrogen interactions at the hypothalamic subcellular level. In *Drug Interactions* (Edited by Morselli P. L., Garattini S. and Cohen S. N.), Raven Press, New York (1974) pp. 151-162.
36. Raynaud J. P. and Moguilewsky M.: Steroid competition for estrogen receptors in the central nervous system. In *Clinical Reproductive Neuroendocrinology* (Edited by Hubinont P. O., L'Hermite M. and Robyn C.), Progress in Reproductive Biology, Vol. 2. S. Karger, Basel (1977) pp. 78-87.
37. Raynaud J. P., Moguilewsky M. and Vannier B.: Influence of rat estradiol binding plasma protein (EBP) on estrogen binding to its receptor and on induced biological responses. In *The Development of Responsiveness to Steroid Hormones* (Edited by Kaye, A. M.), Pergamon Press, Oxford (1979) in press.
38. Moguilewsky M. and Raynaud J. P.: Defeminization: Decreased lordosis response to progesterone in estrogen-primed rats is related to decreased estrogen induction of hypothalamic progestin binding sites.

- 61st Meeting of the Endocrine Soc. (1979) Abstract no. 812.
39. Blaustein J. D. and Wade G. N.: Progesterin binding by brain and pituitary cell nuclei and female rat sexual behavior. *Brain Res.* **140** (1978) 360–367.
 40. Asselin J., Labrie F., Gourdeau Y., Bonne C. and Raynaud J. P.: Binding of (3H)methyltrienolone (R 1881) in rat prostate and human benign prostatic hypertrophy (BPH). *Steroids* **28** (1976) 449–459.
 41. Asselin J., Melancon R., Gourdeau Y., Labrie F., Bonne C. and Raynaud J. P.: Specific binding of [3H]-methyltrienolone to both progestin and androgen binding components in human benign prostatic hypertrophy (BPH). *J. steroid Biochem.* **10** (1979) 483–486.
 42. Zava D. T., Landrum B., Horwitz K. B. and McGuire W. L.: Androgen receptor assay with [3H]methyltrienolone (R1881) in the presence of progesterone receptors. *Endocrinology* **104** (1979) 1007–1012.
 43. Bouton M. M. and Raynaud J. P.: The relevance of interaction kinetics in determining biological response. *Endocrinology* **105** (1979) 509–515.

DISCUSSION

Rousseau. Dr Raynaud, based on diffraction studies you have suggested that the relative lack of specificity of a steroid for a particular receptor is due to a greater flexibility of that steroid. Now, when you consider the nonsteroidal analogues of active steroid hormones for which you expect an even greater flexibility, would you not predict and actually find very little specificity?

Raynaud. If you are referring to compounds such as diethylstilbestrol, then M. Hospital has shown that, in the crystal, one of the conformations of DES fits perfectly that of estradiol as regards the relative positions of the hydroxyl groups. This is somewhat different from highly flexible steroid molecules, in particular 3-keto- Δ^4 steroids, which may approximate the conformation of one or more endogenous hormones but which do not necessarily show the same good fit. As regards non-steroid anti-androgens such as flutamide and RU 23908, these compounds compete appreciably for androgen binding after very short incubation times only. The association with the receptor is probably extremely fast and the dissociation even faster, thus explaining the need for large doses of these compounds for *in vivo* activity. As far as we know, they do not interact with receptors other than the androgen receptor under standard incubation conditions. Maybe they form even more fleeting complexes with these receptors which are as yet undetected and which do not apparently give rise to any known anti-hormonal activity.

Schrader. When you did the binding studies you compared the ability of the various agents to bind as a function of time. You expressed the information as a percent of relative binding ability. Was the number of receptor sites a constant throughout the experiment?

Raynaud. We used a radioligand concentration approaching saturation. The binding experiments were performed under conditions where about 70 to 80% of the binding sites were saturated, thus limiting degradation.

Schrader. So the receptor concentration that you are looking at is essentially constant throughout the assay?

Raynaud. Yes, it is. Furthermore, in experiments where one wishes to establish whether a compound can interact at all with a receptor at a given temperature, it is necessary that the incubation time should be as short as possible; this leaves little time for degradation. The incubation time will depend upon the time required to reach equilibrium and thus upon the receptor. For the progesterone receptor, Scatchard analysis does not give a straight line but an ascending, then descending curve; therefore, for this receptor, incubation time may be a limiting factor. The androgen and estrogen receptors are much more stable. For the mineralocorticoid and glucocorticoid receptors, in my opinion, much of the available data has to be revised on account the cross-specificity of available radioligands, using new specific compounds to eliminate the interference of one binder. The data of our screening system should not be considered more than indicative and for more information improvements to the system are required. However any improvement is also liable to defeat the purpose of the

system by rendering it much more complicated. May I remind you that the basic point I have tried to put across is that it is possible, on the basis of very simple experiments under well-chosen conditions, to classify large numbers of available compounds according to specificity and activity and eventually to design new compounds of predetermined specificity and activity.

Schrader. A variety of laboratories have shown that the equilibrium constant and the ratios of the rate constants for hormone binding do not agree with each other. This suggests a possibility, which has been proposed by a number of investigators, that the hormone binding mechanism on the receptor protein is not a simple one. One interpretation has been that there are two hormone binding sites on the receptor. One explanation for some of your binding studies might be that in fact the competitors you are using are binding to a different hormone binding site from that occupied by the radioactive ligand. There are two ways of testing that idea. One is to prepare these interesting compounds as radioactive ligands and to show that the answer you get by direct binding studies is identical to what you get by competition. The other is to attempt to map the steroid specificity of the competitor binding sites and compare it with the authentic ligand binding site. Have you done either of those two approaches?

Raynaud. Yes, the second one. The trouble is that we have too much information and not always the right information i.e. on the most informative compounds. It is very difficult to assess the influence of one, let alone several, substituents on a structure unless one studies a well-defined series of compounds differing by the relevant substitutions only. And even then, although we can measure how a substituent enhances or impedes binding in a given molecule, we can only predict with difficulty whether its introduction into another molecule will have the same effect, since from time to time a substituent completely changes the shape of a molecule. We have studied in some detail a series of about 40 homologous steroids differing by their degree of unsaturation (a series of monoenes and the corresponding dienes and trienes) and thus have been able to deduce the effect of this type of unsaturation on steroid flexibility and specificity.

Schrader. Haven't you been interpreting this as indicating the interactions of these various compounds occur at one binding site on the protein? I don't think that there is any proof that that is going on.

Raynaud. All we can say is that we have a competitive effect versus a given ligand which seems to follow in a first approximation, the law of mass action for binding to a single site. We have as yet no real evidence for multiple sites. That the ratio of the rates of association and dissociation does not equal the intrinsic dissociation constant for the estrogen receptor for instance, does not disprove the applicability of the law of mass action but can be explained by the very long time required to reach equilibrium for a direct measurement of the intrinsic dissociation constant.

Rochefort. I would like to add that what you propose,

Jean-Pierre, about the relationship between dissociation rate and antagonist activity might not be general. For instance, it does not agree with the estrogen receptor-antiestrogen interactions, as we recently published (Rochefort H., Garcia M. and Borgna J.-L., *Biochem. biophys. Res. Commun.* **88** (1979) 351-357).

Raynaud. I presume that you are referring yet again to the tamoxifen story. In our hands the hydroxylated metabolite of tamoxifen is an estrogen agonist.

Rochefort. I should say that the hydroxy tamoxifen in the MCF₇ cells is in fact a full antagonist (Westley B. *et al.*, in preparation) while its affinity for the estrogen receptor is high.

Schrader. Well, in other systems, Henri, it is an agonist.

Raynaud. In any case, this metabolite only represents about 1% of the parent compound and in my opinion can in no way be responsible for the anti-hormonal activity of tamoxifen in animal and human studies.

Rochefort. Finally, I would like to add that androgen, which dissociates very rapidly from the estrogen receptor, appears to be an agonist and not an antagonist in the uterus; so it seems that there are exceptions for the relationship between dissociation rate and antagonist activity.

Setchell. Can I expand on that a bit? We at ICI have looked at hydroxy tamoxifen and find that, as you've just said, it binds well to the receptor at both time periods and therefore would be predicted to be an estrogen. It is an estrogen in the mouse, but then tamoxifen is an estrogen in the mouse. In the rat, it behaves as an anti-estrogen, and I think there are other exceptions to the general rule that you have made, apart from this one. The second exception is that the *cis* form of tamoxifen is the pure tri-isomer, but the *cis* form behaves in your system exactly as an anti-estrogen ought to behave in that it binds fairly well over short periods, but you lose binding over longer time, and yet this compound has no demonstrable anti-estrogenic activity in the rat, the mouse or the rabbit.

Finally, what I think you've said fails to take into account the species differences that I alluded to earlier. But we've looked at both the mouse and rat uterine estrogen receptors and we get the same effects with tamoxifen—a shift in relative binding in both species. Yet in the mouse, tamoxifen is a full agonist with no demonstrable antagonism. In the rat, in our view, it is an antagonist. Before Dr Clark worries about what we mean by antagonist, if he's talking about the acute hyperemic response, all the compounds we've been talking about this morning are agonists; if he's talking about the ability to produce progesterone receptor in the rat uterus, then the compounds I've spoken about are agonists. But the full response, that is the proliferative response, the mitotic response, tamoxifen just does not give that. It causes cell hypertrophy, but it fails to cause hyperplasia with chlomophine, and what's more it will completely block the hyperplastic response induced by estradiol. So I have considerable reservations about the theory, although we were extremely attracted to it initially

and I had to overcome some prejudices I had about the mechanism of action to abandon it, but abandon it we have.

Clark. I think the point about species differences is well taken and all of us who would like to generalize about hormone action often ignore the species difference. The shrew, an insectivore, is a good example. These animals have very few estrogen receptors in their uterus and do not depend on estrogen for growth of the uterus. Uterine growth is independent of estrogens.

Setchell. I think that the problem you have is a pharmacokinetic one.

Siiteri. The question of species differences has come up and I think that it is important to keep in mind differences observed in the mouse or rat or whatever you are using from the human. You have touched upon the problems we have all had trying to understand the androgen receptor of the human prostate. I wasn't sure of what you said finally. Do you think there really is a progesterone receptor or is it a sloppy androgen receptor?

Raynaud. At the Prostatic Cancer Workshop held in Amsterdam this April, it was clear that all who have studied benign human prostate hypertrophy have found, like us, a progestin-like receptor in the prostate. According to our specificity results, this receptor has the characteristics of a progesterone receptor, but we have no further data. I would personally still call it a progestin-like receptor, until further experimental evidence is available, although many at the workshop considered it a true progesterone receptor. In any case, it is not an artefact and its presence remains to be explained.

Siiteri. I believe that the observations are correct but from a physiological point of view there is hardly any biological reason to have a progesterone receptor in the prostate.

Raynaud. It seems that the stromal cells which contain this progestin-like receptor also contain estrogen receptor. Thus we may be in a situation where estrogens are inducing the synthesis of the progestin receptor in estrogen-sensitive prostate cells.

Siiteri. But functionally speaking there is no need for a progesterone receptor. It may be a rudimentary survivor from evolution, like the appendix. The other thing that bothered me a little in the first part of your talk concerned D-norgestrol being an androgen on the basis of a mouse assay. Norgestrol binds very avidly to SBG and the reason it's androgenic in women is probably because it displaces testosterone and raises the free testosterone to very high levels. Conclusion from interactions at the receptor level in rodents may not explain effects in humans.

Raynaud. Admittedly competition for specific plasma proteins can raise endogenous hormone concentrations in the case of certain progestins, but in our screening system we have detected androgen binding for progestins which do not compete for SBP binding and which are nevertheless androgenic. Progesterone itself has slight androgenic activity demonstrated in both animal and human models.